

Initial Project Planning Template

Date	14 July 2026
Team ID	SWTID-2026-7049
Project Name	Emotion Detection & Learning Support Engine (AI Learning Assistant)
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation

Reconstructed from the project's real commit history (46 commits, 5–14 July 2026), organized by epic. Each sprint below corresponds to a completed epic in that history, ending at "Complete Epic 6 T2: deployment readiness" — the last commit before an unrelated, exploratory run of bug-fix iterations (Hugging Face model-loading retries, OpenRouter timeout tuning) that aren't modeled as separate backlog items here.

Sprint	Epic	USN	User Story / Task	SP	Priority	Team Members	Start	End
Sprint 1	Development Environment	USN-1	As a developer, I can set up a Python 3.12 virtual environment with TensorFlow and CUDA-enabled PyTorch so both frameworks are available for training.	3	High	Aditya Sinha	05-Jul	05-Jul
Sprint 1	Development Environment	USN-2	As a developer, I can verify GPU acceleration and confirm TensorFlow/PyTorch can coexist safely in the environment before writing any model code.	2	High	Aditya Sinha	05-Jul	05-Jul
Sprint 2	Dataset Preparation & Model Training	USN-3	As a developer, I can download and verify GoEmotions, EmpatheticDialogues, and ISEAR so training data is ready and validated.	3	High	Aditya Sinha, Prateek Singh	06-Jul	06-Jul
Sprint 2	Dataset Preparation & Model Training	USN-4	As a developer, I can analyze class distribution and identify that "Bored" is significantly underrepresented, so I can choose an appropriate mitigation strategy.	2	High	Aditya Sinha	06-Jul	06-Jul
Sprint 2	Dataset Preparation & Model Training	USN-5	As a developer, I can train a BiLSTM classifier with class weighting so all five emotion classes are represented without oversampling or synthetic data.	5	High	Aditya Sinha	06-Jul	06-Jul
Sprint 2	Dataset Preparation & Model Training	USN-6	As a developer, I can apply domain-adaptive fine-tuning before the main BERT training pass so the model transfers better to student-written text.	5	High	Aditya Sinha	06-Jul	06-Jul
Sprint 2	Dataset Preparation & Model Training	USN-7	As a developer, I can fine-tune a BERT classifier on the merged dataset so I have a stronger second model to compare against BiLSTM.	5	High	Aditya Sinha	06-Jul	06-Jul
Sprint 2	Dataset Preparation & Model Training	USN-8	As a developer, I can verify both trained models load correctly and agree on label ordering before integrating them into the app.	2	High	Aditya Sinha, Khushboo Goyal	06-Jul	06-Jul

Sprint	Epic	USN	User Story / Task	SP	Priority	Team Members	Start	End
Sprint 3	Inference & Persistence Pipeline	USN-9	As a developer, I can clean and enhance input text with keyword-aware preprocessing so both models receive consistent input.	3	High	Aditya Sinha	06-Jul	06-Jul
Sprint 3	Inference & Persistence Pipeline	USN-10	As a developer, I can wrap the BiLSTM model behind a clean 5-class prediction interface.	2	High	Aditya Sinha	06-Jul	06-Jul
Sprint 3	Inference & Persistence Pipeline	USN-11	As a developer, I can wrap the BERT model with class weighting and keyword adjustments behind the same interface.	3	High	Aditya Sinha	06-Jul	06-Jul
Sprint 3	Inference & Persistence Pipeline	USN-12	As a student user, I want the system to recognize when I'm feeling more than one emotion at once, so the response doesn't oversimplify how I feel.	3	Medium	Aditya Sinha	06-Jul	06-Jul
Sprint 3	Inference & Persistence Pipeline	USN-13	As a developer, I can standardize both models' outputs into one schema so downstream code doesn't need to special-case either model.	2	Medium	Aditya Sinha	06-Jul	06-Jul
Sprint 3	Inference & Persistence Pipeline	USN-14	As a developer, I can cache model loading and persist interactions to CSV so repeated runs are fast and history isn't lost within a session.	3	Medium	Aditya Sinha, Priyanshu Sharma	06-Jul	06-Jul
Sprint 4	AI Response Generation	USN-15	As a student user, I can describe my field and problem and have a Gemini prompt built from my detected emotion and confidence.	3	High	Aditya Sinha	06-Jul	09-Jul
Sprint 4	AI Response Generation	USN-16	As a student user, I want a generated response that's empathetic and specific to my field, with a safe template fallback if AI generation fails.	5	High	Aditya Sinha, Priyanshu Sharma	09-Jul	09-Jul
Sprint 4	AI Response Generation	USN-17	As a student user, when I toggle AI responses on or off, I want the response to regenerate correctly while the underlying model scores stay in sync.	3	Medium	Aditya Sinha	09-Jul	09-Jul
Sprint 4	AI Response Generation	USN-18	As a student user, I want my session's interaction history retained so patterns across multiple problems can be tracked.	3	Medium	Aditya Sinha	09-Jul	09-Jul
Sprint 4	AI Response Generation	USN-19	As a developer, I can build a data-driven decision engine that decides which model to trust and why, based on real validation analysis rather than a fixed rule. (Self-directed addition, outside the original epic plan.)	5	High	Aditya Sinha	09-Jul	09-Jul
Sprint 5	Streamlit UI	USN-20	As a student user, I want a clean single-page layout with persistent session state so my results don't disappear on every interaction.	3	High	Aditya Sinha, Parteek Jangir	13-Jul	13-Jul
Sprint 5	Streamlit UI	USN-21	As a student user, I want to see field selection, problem input, model comparison, mixed-emotion display, confidence bars, and analytics tabs, all in one flow.	5	High	Aditya Sinha	13-Jul	13-Jul

Sprint	Epic	USN	User Story / Task	SP	Priority	Team Members	Start	End
Sprint 5	Streamlit UI	USN-22	As a student user, I want clear form controls, a working clear/reset action, loading spinners, and graceful error handling.	3	Medium	Aditya Sinha, Khushboo Goyal	13-Jul	13-Jul
Sprint 5	Streamlit UI	USN-23	As a student user, I want to see my emotional trends visualized as interactive charts that don't crash the app on repeated use.	3	Medium	Aditya Sinha	13-Jul	13-Jul
Sprint 6	Validation & Deployment Readiness	USN-24	As a QA reviewer, I want an automated end-to-end test suite plus a manual click-through pass, so the full pipeline is verified before deployment.	5	High	Aditya Sinha, Prateek Singh	14-Jul	14-Jul
Sprint 6	Validation & Deployment Readiness	USN-25	As a team, we want the app deployed publicly (even in a reduced/degraded mode) so evaluators can access it without a local setup.	5	High	Aditya Sinha, Parteek Jangir	14-Jul	14-Jul

Note on Team Members column: the repository's commit history is authored under a single account (Aditya Sinha). Task ownership shown above reflects a reasonable division of labour across the team's five members by role (training/ML, UI, testing, deployment); adjust the names per row if your team's actual internal division differs.